

$$\textcircled{1} a) \left. \begin{aligned} \frac{\partial F_1}{\partial y} &= 2y \cos(xy) - xy^2 \sin(xy) + 2\alpha^2 x \\ \frac{\partial F_2}{\partial x} &= y \cos(xy) + y \cos(xy) - xy^2 \sin(xy) + 2\alpha x \end{aligned} \right\} \Rightarrow$$

$$\frac{\partial F_2}{\partial x} = \frac{\partial F_1}{\partial y}$$

$$\Rightarrow 2\alpha^2 x = 2\alpha x \quad \forall x \in \mathbb{R} \quad \dots \quad x(1-\alpha) = 0 \quad \forall x \in \mathbb{R}$$

Tedy $\alpha = 0$ nebo $\alpha = 1$.

b) Hledáme potenciál f vezt. pole F v případě $\alpha = 0$. Tedy f je řeš. rovnice:

$$\frac{\partial f}{\partial x} = y^2 \cos(xy) \quad (1)$$

$$\frac{\partial f}{\partial y} = \sin(xy) + xy \cos(xy) + y^2 \quad (2)$$

$$(1) \Rightarrow f(x, y) = y \sin(xy) + C(y)$$

Dosažením do (2) máme

$$\cancel{\sin(xy)} + \cancel{xy \cos(xy)} + C'(y) = \cancel{\sin(xy)} + \cancel{xy \cos(xy)} + y^2$$

$$C'(y) = y^2 \dots C(y) = \frac{y^3}{3} + K, \quad \text{ kde } K \in \mathbb{R}$$

Protože $10 = f(0, 3) = C(3) = 9 + K$, je $K = 1$.

$$\text{Tedy } f(x, y) = y \sin(xy) + \frac{y^3}{3} + 1.$$

② i) Podstatné body v int M :

$\frac{\partial f}{\partial x} = e^y = 0 \dots$ nebo splnit pro každé $y \in \mathbb{R} \Rightarrow$ v int M neleží žádný bod extrémů f na M .

ii) Podstatné body v ∂M :

$$\begin{aligned} e^y + \lambda 2x &= 0 \dots \lambda = -\frac{e^y}{2x} \quad (x \neq 0, \text{ neboť } \\ x e^y + 2\lambda y &= 0 \quad \text{v případě } x=0 \text{ máme} \\ x^2 + y^2 &= 2 \quad e^y = 0, \text{ což není možné}). \end{aligned}$$

$$\text{Tedy } x e^y + 2\left(-\frac{e^y}{2x}\right)y = 0 \dots e^y\left(x - \frac{y}{x}\right) = 0$$

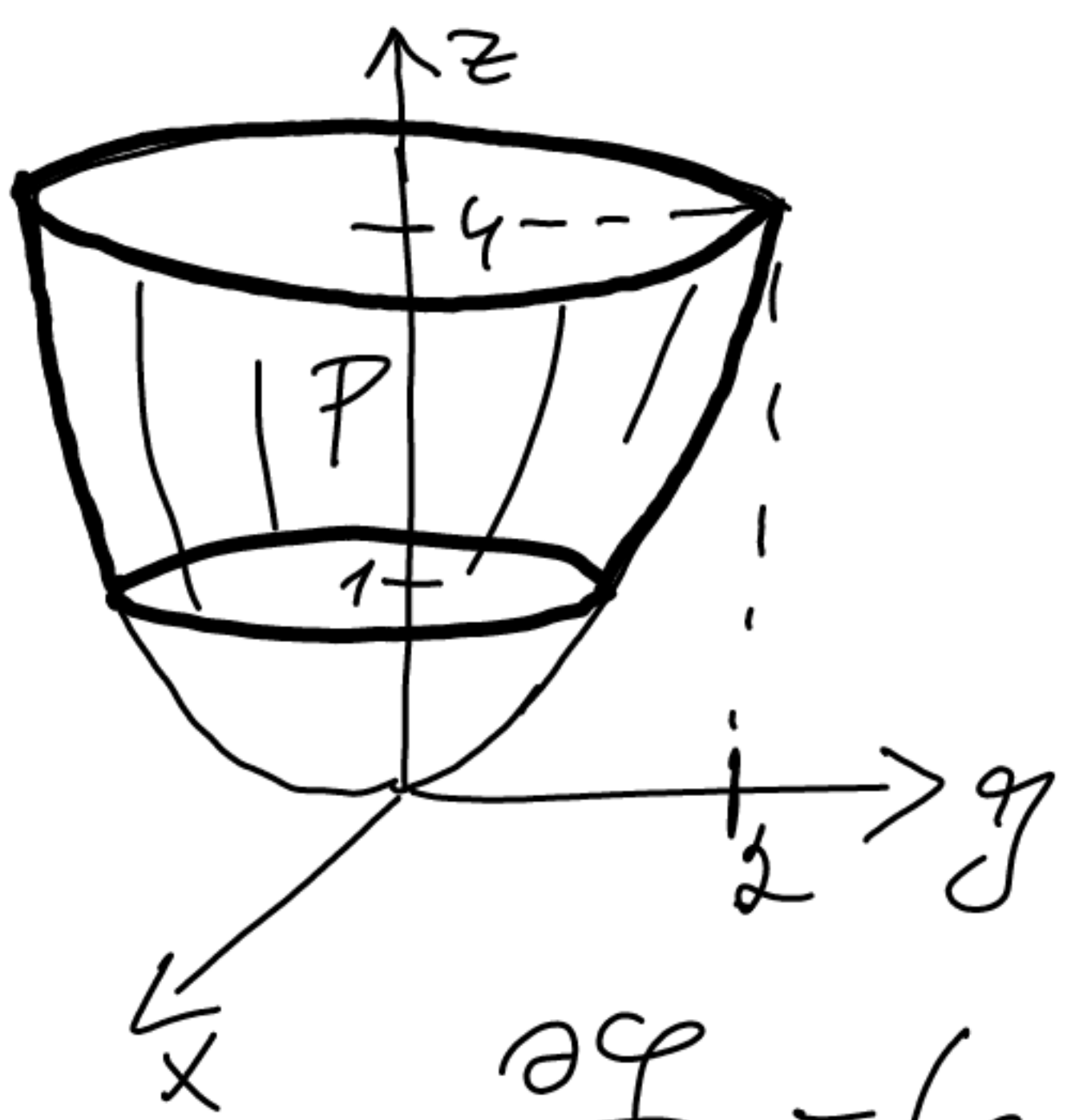
$$\text{Odtud } x = \frac{y}{x} \dots x^2 = y$$

$$\begin{aligned} \text{Proto } y + y^2 &= 2 \dots y^2 + y - 2 = 0 \\ & \quad (y+2)(y-1) = 0 \quad \begin{cases} y = -2 \\ y = 1 \end{cases} \end{aligned}$$

$y = -2$ vyloučíme, neboť $y = x^2 \geq 0$. Tedy podstatné body v ∂M jsou $(-1, 1)$ a $(1, 1)$.

$$\left. \begin{aligned} f(-1, 1) &= -e \\ f(1, 1) &= e \end{aligned} \right\} \Rightarrow \begin{aligned} (-1, 1) &\text{ je bod minima } f \text{ na } M \\ (1, 1) &\text{ je bod maxima } f \text{ na } M \end{aligned}$$

③



$$\varphi(u, v) = (u \cos v, u \sin v, u^2)$$

$$u \in [1, 2]$$

$$v \in [0, 2\pi]$$

$$\left. \begin{aligned} \frac{\partial \varphi}{\partial u} &= (\cos v, \sin v, 2u) \\ \frac{\partial \varphi}{\partial v} &= (-u \sin v, u \cos v, 0) \end{aligned} \right\} \Rightarrow$$

$$\Rightarrow \frac{\partial \varphi}{\partial u} \times \frac{\partial \varphi}{\partial v} = \begin{vmatrix} e_1 & e_2 & e_3 \\ \cos v & \sin v & 2u \\ -u \sin v & u \cos v & 0 \end{vmatrix} =$$

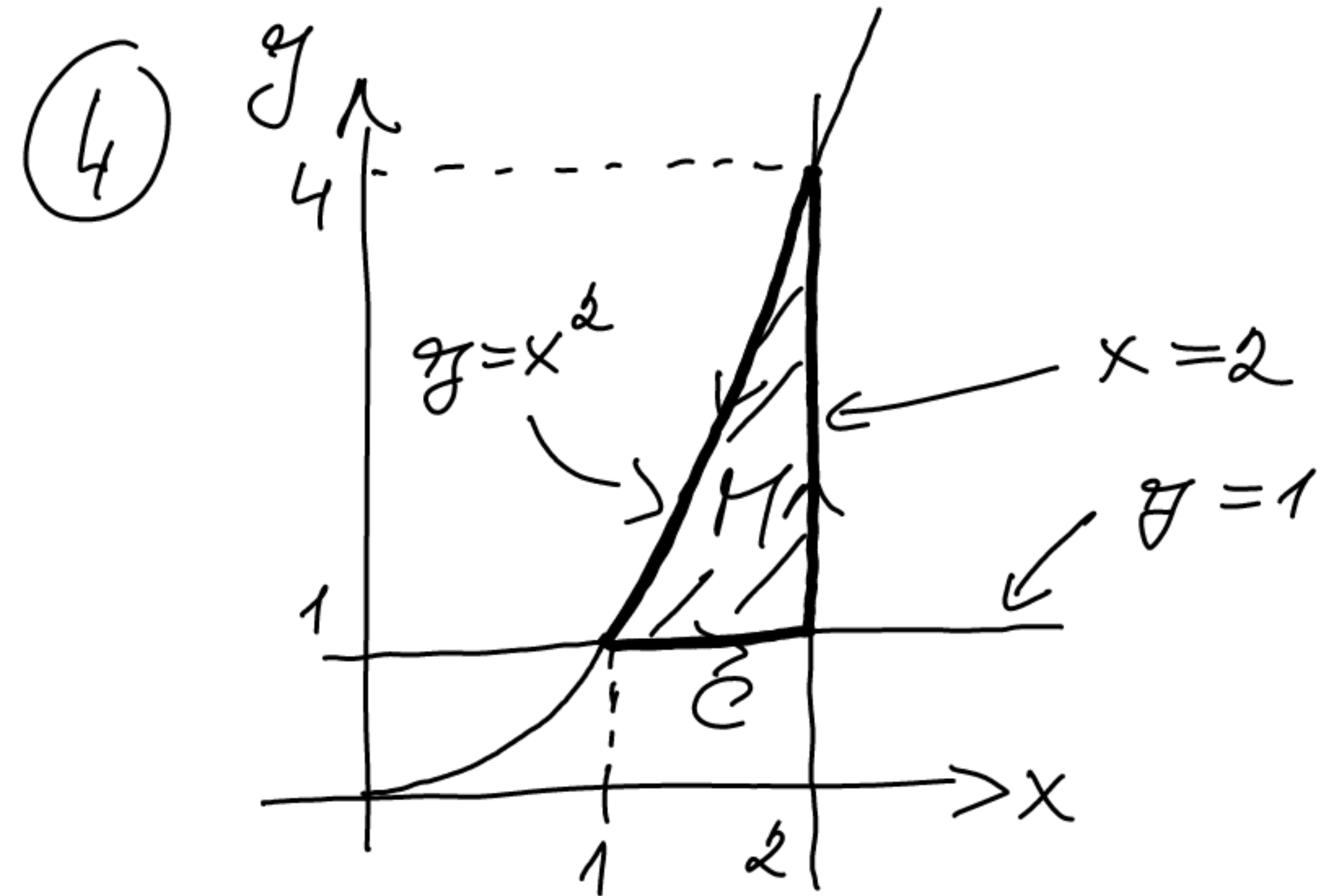
$$= (-2u^2 \cos v, -2u^2 \sin v, u) \Rightarrow$$

$$\Rightarrow \left\| \frac{\partial \varphi}{\partial u} \times \frac{\partial \varphi}{\partial v} \right\| = \sqrt{4u^4 + u^2} = u \sqrt{4u^2 + 1}$$

$$\text{obeah}(P) = \int_P 1 \, d\sigma = \int_0^{2\pi} \int_1^2 u \sqrt{4u^2 + 1} \, du \, dv$$

$$= \left| \begin{aligned} 4u^2 + 1 &= t \\ 8u \, du &= dt \\ 1 \rightsquigarrow 5, 2 \rightsquigarrow 17 \end{aligned} \right| = \frac{2\pi}{8} \int_5^{17} \sqrt{t} \, dt$$

$$= \frac{2\pi}{8} \left[\frac{t^{3/2}}{\frac{3}{2}} \right]_5^{17} = \frac{\pi}{6} \left(17^{3/2} - 5^{3/2} \right)$$



$$\int_C F(x, y) \cdot ds = \int \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} d\vec{r}_2$$

$$= \int_1^2 3 + \frac{x}{y^2} d\vec{r}_2 = \int_1^2 \int_1^{x^2} 3 + \frac{x}{y^2} dy dx$$

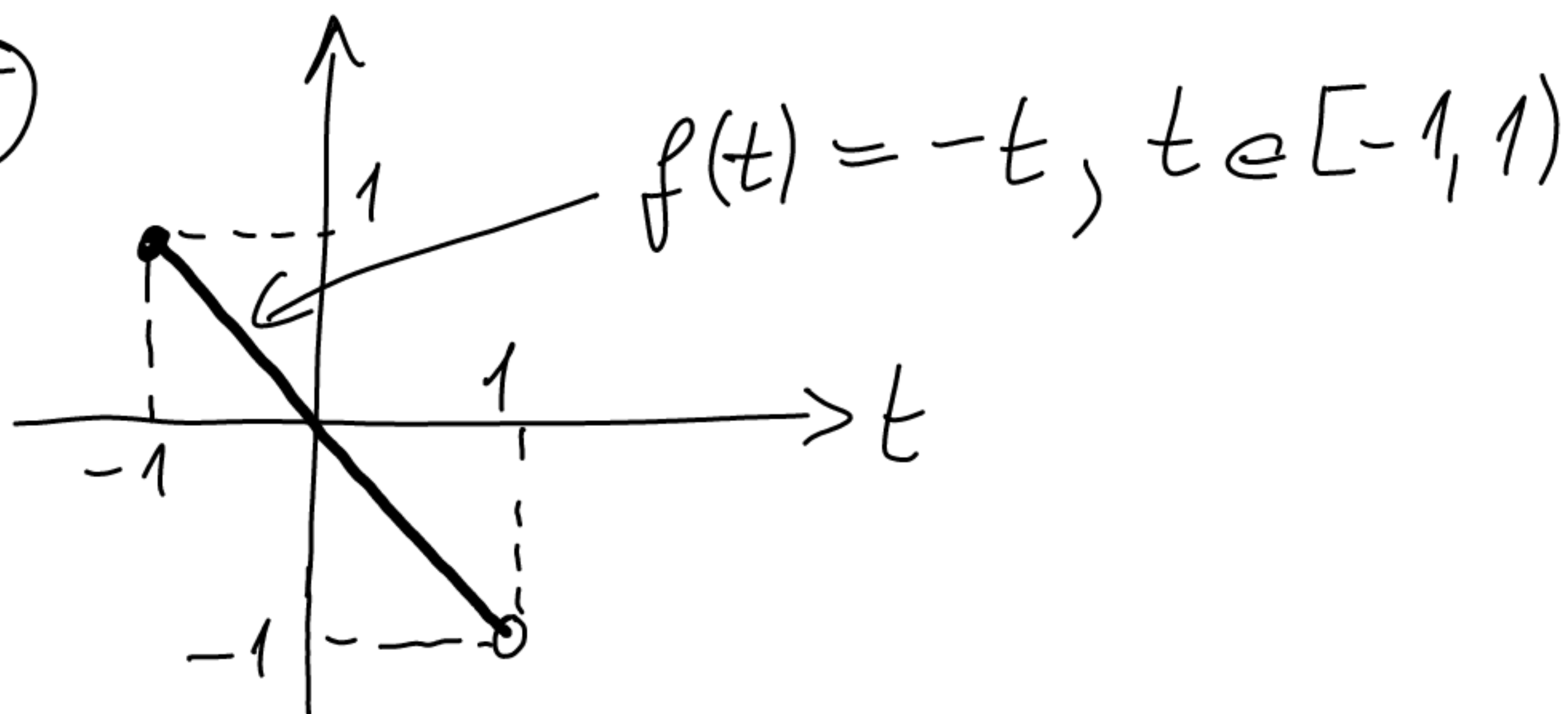
$$= \int_1^2 3x^2 - 3 + x \left[-\frac{1}{y} \right]_1^{x^2} dx$$

$$= \int_1^2 3x^2 - 3 - \frac{1}{x} + x dx$$

$$= \left[x^3 - 3x - \ln x + \frac{x^2}{2} \right]_1^2 = 4 - \ln 2 - \left(-2 + \frac{1}{2} \right)$$

$$= \frac{11}{2} - \ln 2$$

⑤



f je ličebná na $(-1, 1) \Rightarrow a_n = 0$

$$\begin{aligned} b_n &= \int_{-1}^1 -t \sin(\pi n t) dt = -2 \int_0^1 t \sin(\pi n t) dt \\ &= -2 \left\{ \left[-\frac{t \cos(\pi n t)}{\pi n} \right]_0^1 - \int_0^1 -\frac{\cos(\pi n t)}{\pi n} dt \right\} \\ &= \frac{2(-1)^n}{\pi n} - 2 \left[\frac{\sin(\pi n t)}{\pi^2 n^2} \right]_0^1 = \frac{2(-1)^n}{\pi n} \end{aligned}$$

Fourierova řada funkce f je

$$\sum_{n=1}^{\infty} \frac{2(-1)^n}{\pi n} \sin(\pi n t).$$

Její součet na $[-1, 1)$ je

$$\tilde{f}_f(t) = \begin{cases} -t, & t \in (-1, 1) \\ 0, & t = -1. \end{cases}$$